

We performed positive steering control experiments to validate the directional behavior of our feature, applying scalar values from 0.069 to 0.693. As predicted, steering the feature positively resulted in a systematic increase in hydrophobicity (GRAVY score) while maintaining structural stability at moderate steering strengths. The modified sequence GRAVY scores increased from -0.36 to 1.53, reflecting enhanced hydrophobic content, while SASA values remained stable ($\leq 3\%$ change) until the highest scalar values where minor structural perturbations emerged.

Additionally, to assess the specificity of identified feature, we performed negative steering experiments on randomly selected features from each Matryoshka group (features 5, 39, 1381, and 7921). Unlike our target feature, random features showed minimal response to steering, with three features showing no changes ($\leq 0.1\%$ variation) across all metrics and one feature (Feature 5) showing only minor variations incomparable to the dramatic shifts observed in our main experiment.

I IMPLEMENTATION DETAILS

I.1 SAE IMPLEMENTATION DETAILS

For each token of a protein sequence, a transformer encoder-based PLM outputs an embedding x in each layer ℓ . To learn meaningful representations, sparsity constraints are imposed on the latent vector z . This sparsity can be achieved either through L1 regularization terms in the loss function or through specialized activation functions that directly restrict the number of non-zero elements. In practice, the number of active (non-zero) elements K satisfies $K \ll d \ll n$, preventing degenerate solutions such as the identity map.

Training and Normalization. During training, we normalize our embeddings to have average unit L2 norm, enabling hyperparameter transfer between layers. This normalization is reversed during inference by scaling the biases according to the procedure detailed in Marks et al. (2024). This approach ensures consistent performance across different layers while maintaining the model’s representational capacity.

I.2 CODE

For all SAE implementations, we use Marks et al. (2024). For our L1 regularized SAEs, we follow Conerly et al. (2024). For TopK, we follow Gao et al. (2024). In Fig 2a, we standardize hyperparameters with an expansion factor of 8x (dict size 20480) and batch size 2048 and use best learning rate.

I.3 DATA CURATION

Following the approach outlined in the InterPLM paper, we first process the UniRef50 FASTA file by iterating over each protein sequence. We remove any sequence that exceeds 1022 tokens, as ESM-2 cannot process longer inputs. From the remaining sequences, we randomly select a predetermined number of proteins to create our working dataset. This dataset is then divided into shards of 1000 sequences each, facilitating efficient handling during training.

I.4 OPTIMIZING THE DATASET AND TRAINING PROCEDURE

After generating embeddings with ESM-2, the activations are stored as tensors representing entire shards. We then use Litdata’s optimize function to iterate through each tensor and split it into fixed-size batches. The optimized dataset achieves speedup by reducing streaming I/O overhead to AWS S3 during training. Additionally, the optimized dataset uses Litdata’s StreamingDataset and StreamingDataLoader where we shuffle the batches during training, while also leveraging PyTorch Lightning for multi-GPU training support.

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